

Module 2: Excel-Based Data Wrangling

1. TRIM, Duplicate, Find & Replace

A. TRIM()

TRIM removes extra spaces from text except single spaces between words.

Sometimes imported data contains unnecessary spaces that cause errors. TRIM cleans them.

Example

```
=TRIM(" Alisha Khan ")
```

Result:

Alisha Khan

Interview Point

TRIM is commonly used during data cleaning.

One-Line Revision

TRIM removes unwanted spaces.

B. Remove Duplicates

Removes repeated records from a dataset.

If the same customer appears 5 times accidentally, Excel can keep only one record.

Example

Ali

Ali

Ali

Sara

After removing duplicates:

Ali

Sara

One-Line Revision

Remove Duplicates eliminates repeated entries.

C. Find & Replace

Searches for specific values and replaces them automatically.

Instead of changing 500 entries manually, Excel does it instantly.

Shortcut

Ctrl + H

Example:

Karachi → KHI

One-Line Revision

Find & Replace updates data in bulk.

2. Case Transformations

LOWER()

Converts text to lowercase.

Example

=LOWER("ALISHA")

Result:

alisha

Revision

LOWER → all small letters.

UPPER()

Converts text to uppercase.

Example

=UPPER("alisha")

Result:

ALISHA

Revision

UPPER → all capital letters.

PROPER()

Capitalizes the first letter of every word.

Example

=PROPER("alisha khan")

Result:

Alisha Khan

Revision

PROPER → professional name formatting.

3. Data Extraction

LEFT()

Extracts characters from the beginning of text.

Example

=LEFT("Pakistan",4)

Result:

Paki

Revision

LEFT extracts from the left side.

RIGHT()

Extracts characters from the end of text.

Example

=RIGHT("Pakistan",4)

Result:

stan

Revision

RIGHT extracts from the right side.

MID()

Extracts text from the middle.

Example

=MID("Pakistan",3,4)

Result:

kist

Revision

MID extracts from the middle.

LEN()

Counts total characters.

Example

=LEN("Pakistan")

Result:

8

Revision

LEN counts characters.

4. Combining Data

& Operator

Joins multiple values.

Example

=A1&" "&B1

Result:

Alisha Khan

Revision

& joins text.

CONCAT()

Combines multiple strings.

Example

=CONCAT(A1,B1)

Revision

CONCAT combines text values.

TEXTJOIN()

Combines text using a chosen separator.

Example

=TEXTJOIN(", ",TRUE,A1:A5)

Result:

Ali, Sara, Ahmed, Zara

Revision

TEXTJOIN joins text with separators.

5. Splitting Columns

Delimiter Method

Splits data using a special character.

Excel breaks text whenever it sees a chosen symbol.

Example

Ali,Khan

Delimiter:

,

Output:

Ali | Khan

Revision

Delimiter splits using symbols.

Fixed Width

Splits data after a specific number of characters.

Example

123456789

Split after 3 digits:

123 | 456789

Revision

Fixed Width splits by position.

6. Flash Fill

Automatically detects patterns and fills data.

Excel watches what you're doing and finishes the work for you.

Example

Alisha Khan

You manually type:

Alisha

Excel predicts the rest.

Shortcut

Ctrl + E

Revision

Flash Fill learns patterns automatically.

7. Merging, Consolidating & Appending Data

Merging

Combining related datasets using a common field.

Example

Customer table + Order table

Using Customer ID.

Revision

Merge = combine datasets using a key.

Consolidating

Combining summaries from multiple sheets.

Example

January + February + March sales.

Revision

Consolidation combines summaries.

Appending

Adding records underneath existing records.

Example

January sales



February sales



March sales

Revision

Append = stack data vertically.

8. Date & Time Functions

TODAY()

Returns today's date.

=TODAY()

Revision:

Returns current date.

NOW()

Returns current date and time.

=NOW()

⚡ Revision:

Returns date + time.

Yesterday

=TODAY()-1

Tomorrow

=TODAY()+1

YEAR()

Extracts year.

=YEAR(A1)

MONTH()

Extracts month.

=MONTH(A1)

DAY()

Extracts day.

=DAY(A1)

DATEDIF()

Calculates difference between dates.

Example

Age calculation.

=DATEDIF(A1,TODAY(),"Y")

Revision:

Measures time difference.

NETWORKDAYS()

Counts working days excluding weekends.

Example

Project duration calculation.

Revision:

Counts business days.

WORKDAY()

Returns future or past working date.

Example

Add 10 working days.

=WORKDAY(A1,10)

Revision:

Calculates working-date deadlines.

Gantt Chart Style Tracking

A visual project timeline showing task durations.

Tracks who does what and when.

Interview Point

Used heavily in project management.

Revision

Gantt charts visualize project schedules.

9. Logical Functions

IF()

Returns different results based on conditions.

Example

=IF(A1>=50,"Pass","Fail")

Revision:

IF checks one condition.

IFS()

Multiple conditions.

=IFS(...)

⚡ Revision:

Multiple IF statements.

SWITCH()

Matches values.

=SWITCH(A1,...)

⚡ Revision:

Alternative to nested IFs.

AND()

All conditions must be TRUE.

=AND(A1>50,B1>50)

⚡ Revision:

Every condition must pass.

OR()

At least one condition must be TRUE.

=OR(A1>50,B1>50)

⚡ Revision:

One condition is enough.

10. Arithmetic Functions

SUM()

Adds values.

=SUM(A1:A10)

⚡ Revision:

SUM adds numbers.

AVERAGE()

Calculates mean.

=AVERAGE(A1:A10)

⚡ Revision:

AVERAGE finds mean.

PRODUCT()

Multiplies numbers.

=PRODUCT(A1:A5)

⚡ Revision:

PRODUCT multiplies values.

11. Lookup Functions

👤 One of the most important Data Analyst topics.

VLOOKUP()

Searches vertically.

=VLOOKUP(...)

⚡ Revision:

Finds values column-wise.

HLOOKUP()

Searches horizontally.

=HLOOKUP(...)

⚡ Revision:

Finds values row-wise.

INDEX()

Returns value from position.

⚡ Revision:

Returns value at a location.

MATCH()

Returns position.

⚡ Revision:

Returns location number.

INDEX + MATCH

🎯 Industry-preferred before XLOOKUP.

MATCH finds the address, INDEX retrieves the value.

⚡ Revision:

Flexible alternative to VLOOKUP.

XLOOKUP()

Modern lookup function replacing VLOOKUP and HLOOKUP.

💡 **Example**

=XLOOKUP(...)



Interview Point

Many companies now prefer XLOOKUP because it's simpler and more powerful.



Revision:

XLOOKUP is the modern lookup solution.

LOOKUP()

Older lookup function.



Revision:

Legacy lookup method.

12. Error Handling

IFERROR()

Replaces formula errors with custom values.



Example

=IFERROR(A1/B1,0)

⚡ Revision:

Handles errors gracefully.

13. Sorting, Filtering & Grouping

Sorting

📌 **Definition**

Arranges data in order.

💡 **Example**

A → Z

Smallest → Largest

⚡ Revision:

Sorting organizes data.

Filter

Displays only required records.

Shortcut

Ctrl + Shift + L

 Revision:

Filter shows selected data.

Advanced Filter

Uses criteria ranges for complex filtering.

Interview Point

Useful for large datasets.

 Revision:

Advanced Filter applies custom conditions.

Group / Ungroup

Collapses and expands rows or columns.

 Revision:

Group organizes large datasets.

Subtotals with Outlines

Automatically calculates grouped summaries.

Example

Sales by Region.

North → Total

South → Total

 Revision:

Subtotals summarize grouped data.